High-Stability Position-Sensorless Control Method for Brushless DC Motors at Low Speed

Tuanjie Li and Jiaxing Zhou

***Abstract*—In order to improve the stability of brushless DC (BLDC) motors at low speed, a high-stability position-sensorless control method is proposed in this paper. Because the back electromotive force (EMF) is very small at low speed, a novel algorithm is proposed to detect the zero crossing point of back EMF accurately. First, the line-to-line back EMF is computed based on the mathematical model of BLDC motors. Then, a low-pass filter (LPF) with alterable cutoff frequency is used to reduce the disturbance of the line-to-line back EMF. Last, the commutation signal is obtained by detecting the zero crossing point (ZCP) of the line-to-line back EMF. However, the commutation signal is delayed by the LPF. For this reason, based on the three-phase back EMF, a novel compensation algorithm including an open loop and a close loop is proposed to compensate for commutation error. Moreover, the speed feedback has a big delay at low speed. According to this, a novel speed calculation algorithm is presented to decrease the delay. Both the simulation and experimental results validate the high stability and reliability of the proposed method.**

***Index Terms*—Back electromotive force (EMF), brushless DC (BLDC) motor, commutation error, high stability, low speed, sensorless control, zero crossing point (ZCP).**

I. Introduction

BRUSHLESS DC (BLDC) motors possess many advantages, such as simple structure, reliable operation, high efficiency, and high power density. Hence, BLDC motors are widely applied in electric vehicles, robots, and medical instruments [1], [2]. BLDC motors require rotor position information for commutation. However, the problems of high cost, big size, and high installation difficulty must be solved when position sensors such as the Hall-effect sensors or encoder are applied. Furthermore, these position sensors are influenced by the environment easily and limit the motors' application.

In recent years, the position-sensorless control has become one of the major focuses in the motor control field. To implement the high stability and reliability at low speed, two main factors need to be focused on. The first is that it is difficult to detect the zero crossing point (ZCP) directly because of the small back electromotive force (EMF). The second is that the delay of commutation signal, which is caused by the low-pass filter (LPF), needs to be compensated effectively.

The direct detection method of back EMF ZCP gets the phase back EMF based on the phase terminal voltages [3]–[6]. The LPF is used to eliminate the noise introduced by these methods; meanwhile, it brings the phase delay. Furthermore, the ZCP cannot be detected accurately at low speed because of the small phase back EMF and big measured noise. The back EMF integration method [7], [8] extracted the commutation signal through integrating the back EMF. However, this method accumulates error at low speed. The virtual third harmonic back EMF method [9], [10] was also not suitable for low speed because of the large noise disturbance and integral error. The phase back EMF can be obtained based on the mathematical model [11], [12], but the dynamic and inherent errors of model introduce the uncertain deviation. The ZCP detection methods based on the F function [13] and G function [14] cannot avoid the commutation error, which is caused by the LPF and the mathematical model. Therefore, in order to achieve the accurate commutation, an appropriate compensation algorithm must be adopted.

In [15], a proportional-intergral (PI) controller based on phase back EMF was used to compensate for the existing commutation error automatically. However, it requires a Hall-effect sensor to calibrate the initial compensation value. In addition, due to the fixed cutoff frequency of LPF and discontinuous initial compensation value, the commutation signal has a large deviation at low speed. In [16] and [17], a dynamic compensation method based on phase current was proposed. But it is not suitable to the situation that the phase current is small or changeable.

This paper presents a position-sensorless control method for BLDC at low speed. There are three features.

1) This method based on the mathematical model has several advantages including a novel LPF with alterable cutoff frequency and a ZCP detection algorithm based on G function.
2) Both dynamic open-loop and closed-loop compensation algorithms are used to implement precise compensation.
3) The novel speed calculation algorithm can increase the frequency of speed update and improve the system response rate and stability.

The rest of this paper is organized as follows. Section II describes a ZCP detection algorithm and a speed calculation method. A novel compensation algorithm is proposed in Section III. In Section IV, the simulation and experimental

Manuscript received January 29, 2018; revised April 10, 2018, May 27, 2018, and July 12, 2018; accepted July 23, 2018. Date of publication August 5, 2018; date of current version March 29, 2019. Recommended for publication by Associate Editor B. G. Fernandes. *(Corresponding author: Tuanjie Li.)*

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Digital Object Identifier 10.1109/TPEL.2018.2863735

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results are shown to validate the proposed method. Section V gives some primary conclusions.

II. ZCP Detection Algorithm Based on the Mathematical Model

A. Mathematical Model of BLDC Motors

The BLDC motor has three phases and the angle of each two phases is 120°. The terminal voltage of phase A is [14]

$$V_a = R_a i_a + \frac{d}{dt}(L_s i_a + L_m i_b + L_m i_c) + \frac{d\lambda_{ar}(\theta)}{dt} + V_n \tag{1}$$

where V_a is the terminal voltage of phase A, R_a is the resistance, i_a, i_b, and i_c are the three-phase currents, respectively, L_s is the self-inductance, and L_m is the mutual-inductance, $\lambda_{ar}(\theta)$ is the flux linkage that is a function of rotor position, and V_n is the neutral voltage. The units of voltage, current, inductance, angle, and flux linkage are V, A, H, rad, and Wb, respectively.

The three-phase resistances and currents of balanced wye-connected BLDC motors always obey the following equations:

$$R_a = R_b = R_c = R \tag{2}$$

$$i_a + i_b + i_c = 0. \tag{3}$$

The phase inductance under balanced conditions is defined as $L = L_s - L_m$, and the back EMF of phase A is defined as $E_a = \frac{d\lambda_{ar}(\theta)}{dt}$. Then, substituting (2) and (3) into (1) gives

$$V_a = Ri_a + L\frac{di_a}{dt} + E_a + V_n. \tag{4}$$

It is easy to know the back EMF of phase A is

$$E_a = V_a - Ri_a - L\frac{di_a}{dt} - V_n. \tag{5}$$

According to (5), the back EMF of phases B and C can be derived as

$$E_b = V_b - Ri_b - L\frac{di_b}{dt} - V_n \tag{6}$$

$$E_c = V_c - Ri_c - L\frac{di_c}{dt} - V_n \tag{7}$$

where V_b and V_c are the terminal voltages of phases B and C, respectively.

The line-to-line voltages equations are described as

$$\begin{bmatrix} E_{ab} \\ E_{bc} \\ E_{ca} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} E_a \\ E_b \\ E_c \end{bmatrix}. \tag{8}$$

Substituting (5)–(7) into (8), we have

$$\begin{bmatrix} E_{ab} \\ E_{bc} \\ E_{ca} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} - \begin{bmatrix} R & -R & 0 \\ 0 & R & -R \\ -R & 0 & R \end{bmatrix} \times \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} - \begin{bmatrix} L & -L & 0 \\ 0 & L & -L \\ -L & 0 & L \end{bmatrix} \begin{bmatrix} di_a \\ di_b \\ di_c \end{bmatrix} \frac{1}{dt}. \tag{9}$$

TABLE I
Relationship Between the Commutation Signal and Hall Signal

Commutation signal	S1	S2	S3	S4	S5	S6
Hall signal	5	1	3	2	6	4

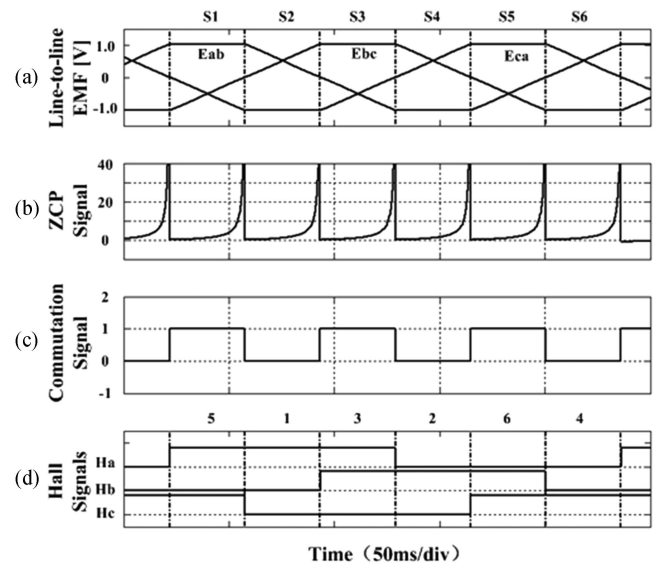

Fig. 1. Proposed ZCP detection method. (a) Line-to-line back EMF. (b) ZCP signal. (c) Commutation signal. (d) Hall signals.

B. ZCP Detection Algorithm

The relationship between the commutation signal and Hall sensor signal is shown in Table I. The ZCPs of line-to-line back EMF are the peak points of the ideal ZCP signal, as shown in Fig. 1.

In order to get the accurate ZCP signal, the G function is adopted. It can be seen from Fig. 1 that E_{bc} closes to zero and E_{ca} closes to maximum of its absolute value on the right boundary of S1. So, E_{ca}/E_{bc} tends to infinity and the ZCP signal can be generated by setting the threshold value of E_{ca}/E_{bc}.

Compared with other ZCP detection algorithms, G function can decrease the noise effectively. For example, during the S1 period, if E_{bc} generates a big noise near the left boundary, the signal of E_{ca}/E_{bc} will be influenced slightly because E_{ca} is near to zero. So, the noise will be reduced. In addition, near the right boundary, E_{ca}/E_{bc} will be magnified because E_{ca} is near absolute maximum and E_{bc} is near zero.

The conditions of ZCP detection are described in Table II.

C. Digital LPF Design

Because the measured terminal voltages and phase currents are interfered by noise, an LPF is introduced to eliminate the interference. In order to eliminate the noise at low speed, the conventional methods set an LPF with very small cutoff frequency in general. However, the LPF with small cutoff frequency will cause two serious problems. One is that the phase delay will increase rapidly because of low speed. The

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TABLE II
CONDITIONS OF ZCP DETECTION

Commutation signal	Conditions	ZCP detection
S1	$E_{ab} > 0$	$ZCP = \frac{E_{ca}}{E_{bc}}$
S4	$E_{ab} < 0$	
S2	$E_{ca} > 0$	$ZCP = \frac{E_{bc}}{E_{ab}}$
S5	$E_{ca} < 0$	
S3	$E_{bc} > 0$	$ZCP = \frac{E_{ab}}{E_{ca}}$
S6	$E_{bc} < 0$	

other is that the filtered signal will be distorted easily when the speed increases. In terms of these problems, a digital LPF with alterable cutoff frequency is proposed. Its cutoff frequency increases as the speed increases and decreases as the speed decreases. The function of the digital LPF is

$$\mathrm{LPF} = \frac{\omega_c}{s + \omega_c} = \frac{2\pi f_c}{s + 2\pi f_c} \tag{10}$$

where ω_c is the cutoff angular velocity and f_c is the cutoff frequency, expressed as

$$f_c = k_c f \tag{11}$$

where k_c is the proportional coefficient and $0 < k_c \leq 1$, and f is the frequency of input signal, expressed as

$$f = \frac{\omega}{2\pi}. \tag{12}$$

Substituting (11) and (12) into (10) gives

$$\mathrm{LPF} = \frac{k_c \omega}{s + k_c \omega}. \tag{13}$$

D. Speed Calculation Algorithm

The traditional speed calculation algorithms obtain the speed by calculating the average speed of one electrical angle period. Its update frequency is in accordance with the commutation. So, the calculated speed has high retardation when the speed of motor is low because of the lag of the average speed itself and the low update frequency. According to this, the closed-loop PI controller is invalid under the sudden load change at low speed because of the high delay of speed feedback, and then the motor stops easily. In addition, the delay also has a direct influence on the calculation of LPF's cutoff frequency and the amplitude of the three-phase, which is used for the compensation algorithm. In order to reduce the delay of speed and improve the stability of control system, a novel speed calculation algorithm is presented as follows:

$$\omega_k = \alpha \frac{10}{p \Delta t_k} + (1 - \alpha)\omega_{k-1} \tag{14}$$

where ω_k is the kth speed, ω_{k-1} is the $(k-1)$th speed, Δt_k is the kth 60° period, p is the number of pole pairs, and α is the proportional coefficient and $0 < \alpha \leq 1$.

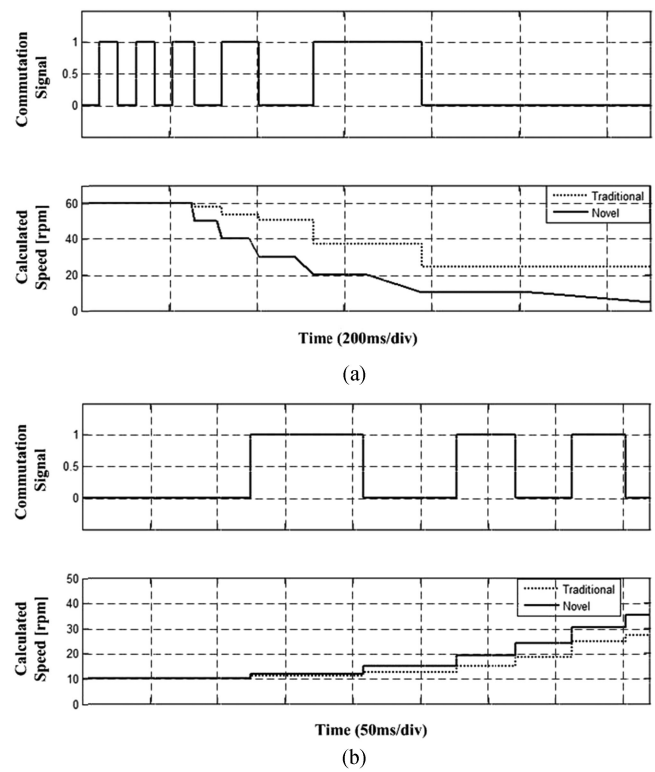

Fig. 2. Calculated speed $(p = 4)$. (a) Deceleration. (b) Acceleration.

A new function is defined as

$$\Delta t = \max\{\Delta t_1, \Delta t_2\}. \tag{15}$$

When $\Delta t > \Delta t_k$ or the commutation takes place, then

$$\begin{cases} \Delta t_k = \Delta t \\ k = k + 1 \end{cases} \tag{16}$$

where Δt_1 is the last commutation period and Δt_2 is the interval from the last commutation to current time.

Because every electrical angle period contains six commutation periods, we define the proportional coefficient $\alpha = 1/6$ to compare with the traditional speed calculation algorithms, and the comparison result of calculated speed is shown in Fig. 2.

From Fig. 2(a), we can see that the proposed speed calculation algorithm can detect the deceleration point in time and improve the response rate effectively at the period of deceleration. From Fig. 2(b), we find that the proposed speed calculation algorithm is better than the traditional ones when the motor accelerates. In addition, by increasing the proportional coefficient α properly, the response rate can be increased further.

III. COMPENSATION ALGORITHM

Because the angle error includes the static error caused by LPF and the dynamic error caused by model error, open-loop and closed-loop compensation algorithms are used to compensate for the static error and dynamic error, respectively. The compensation equation is written as

$$\Delta\theta = \Delta\theta_1 + \Delta\theta_2 \tag{17}$$

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where $\Delta\theta$ is the total compensation angle, and $\Delta\theta_1$ is the open-loop compensation angle, and $\Delta\theta_2$ is the closed-loop compensation angle.

A. Open-Loop Compensation Algorithm

Due to the existence of LPF, the ZCP signal has a long delay. In order to compensate for this delay, the conventional algorithms measure the delay at different speed intervals by a position sensor when the motor has no load. Then, the relationship of the speed and delay is built and used to compensate for the delay. However, the position sensor is very hard to install precisely, and the measured delay is discontinuous. All of these will cause a huge error when the speed changes at low speed. Therefore, a novel open-loop compensation algorithm, which does not rely on position sensor and is continuous, is proposed.

Fig. 1 shows that the waveform of the line-to-line back EMF is similar to sine wave. Hence, the open-loop compensation algorithm, which is used to compensate for the phase delay caused by the LPF, is expressed as

$$\Delta\theta_1 = \tan^{-1}\frac{\omega}{2\pi f_c}. \tag{18}$$

Substituting (11) and (12) into (18) gives

$$\Delta\theta_1 = \tan^{-1}\frac{1}{k_c}. \tag{19}$$

From (19), we can see that $\Delta\theta_1$ is a constant, which can be computed easily.

B. Closed-Loop Compensation Algorithm

Because the waveform of the line-to-line back EMF is not an ideal sine wave and the mathematical model of motors has the inherent and dynamic errors, there are uncertain commutation errors after open-loop compensation. In order to implement precise commutation, these commutation errors must be considered.

When phase A is unexcited, it is easy to know

$$\begin{cases} i_b = -i_c,\ i_a = 0 \\ E_b + E_c = 0. \end{cases} \tag{20}$$

Substituting (20) into (5)–(7) gives

$$\begin{cases} V_a + V_b + V_c = 3V_n + E_a \\ V_a = V_n + E_a. \end{cases} \tag{21}$$

A new function E_{abc} is defined as

$$E_{abc} = \frac{2}{3}E_a = V_a - \frac{V_{tol}}{3} \tag{22}$$

where E_{abc} is the three-phase back EMF and V_{tol} is the sum of terminal voltages, which is

$$V_{tol} = V_a + V_b + V_c. \tag{23}$$

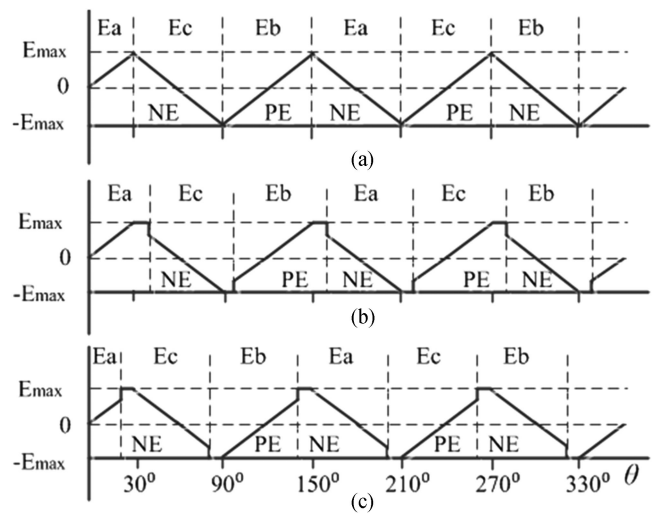

Fig. 3. Three-phase back EMF. (a) Normal commutation. (b) Lagged commutation. (c) Advanced commutation.

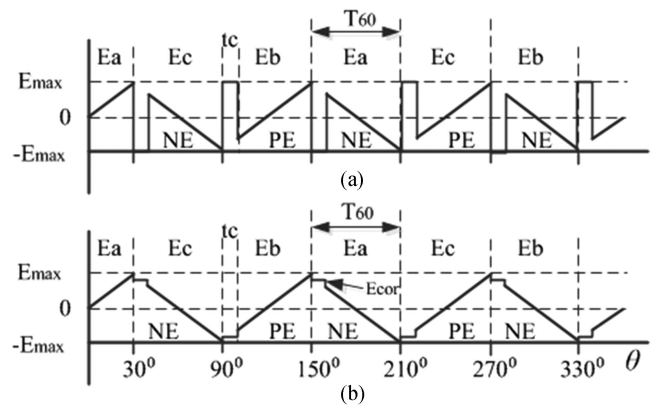

Fig. 4. Fluctuation of three-phase back EMF. (a) Uncorrected. (b) Corrected.

According to (22), when phases B and C are unexcited, respectively, E_{abc} can be derived as

$$E_{abc} = \frac{2}{3}E_b = V_b - \frac{V_{tol}}{3} \tag{24}$$

$$E_{abc} = \frac{2}{3}E_c = V_c - \frac{V_{tol}}{3}. \tag{25}$$

The waveform of E_{abc} is shown in Fig. 3. From Fig. 3(a), we can see that the three-phase back EMF is symmetric in the adjacent interval under the normal commutation. However, under the lagged and advanced commutation, the deviation will be introduced as shown in Fig. 3(b) and (c). The rising stage is defined as positive EMF interval (PE), and the falling stage is defined as negative EMF interval (NE).

Defining the integration of PE and NE as $\int E_{\mathrm{PE}}dt$ and $\int E_{\mathrm{NE}}dt$, respectively, the integration difference can be obtained as $\Delta E = \int E_{\mathrm{PE}}dt - \int E_{\mathrm{NE}}dt$. From Fig. 3, we can find the relationship between ΔE and $\Delta\theta_2$ as follows:

$$\begin{cases} \Delta\theta_2 > 0,\ \Delta E > 0 \\ \Delta\theta_2 = 0,\ \Delta E = 0 \\ \Delta\theta_2 < 0,\ \Delta E < 0. \end{cases} \tag{26}$$

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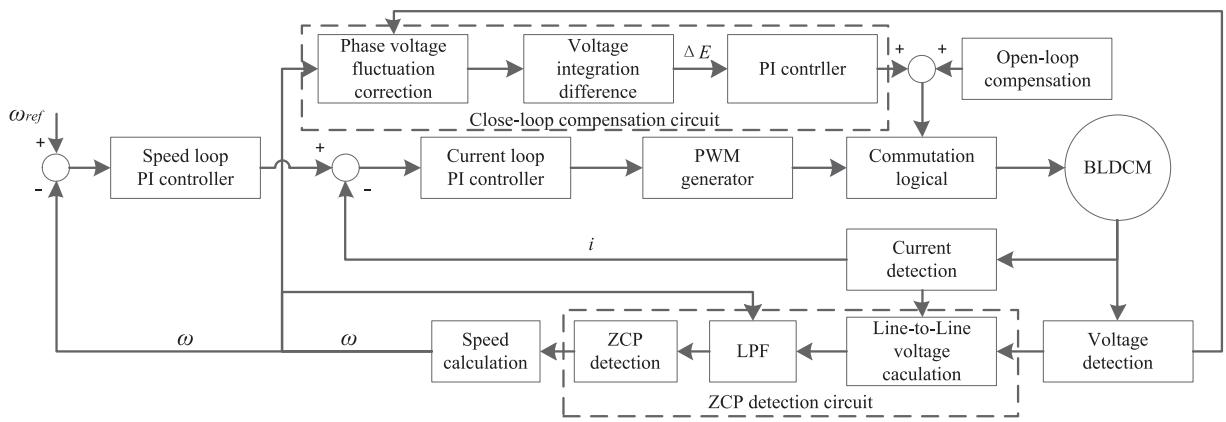

Fig. 5. Speed control system.

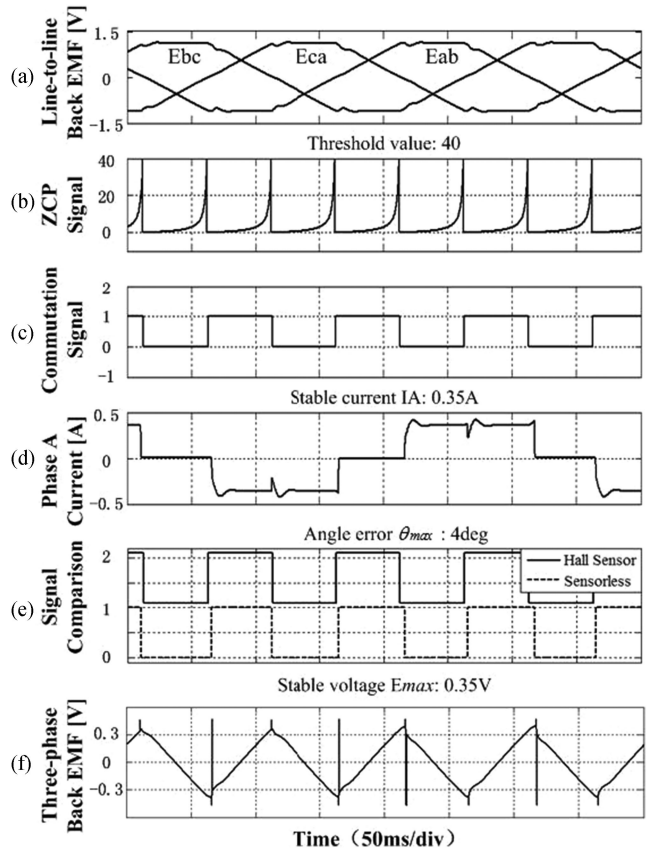

Fig. 6. Simulation results at 50 r/min. (a) Line-to-line back EMF. (b) ZCP signal. (c) Actual commutation signal. (d) Phase A current. (e) Hall sensor signal and actual commutation signal. (f) Corrected three-phase back EMF.

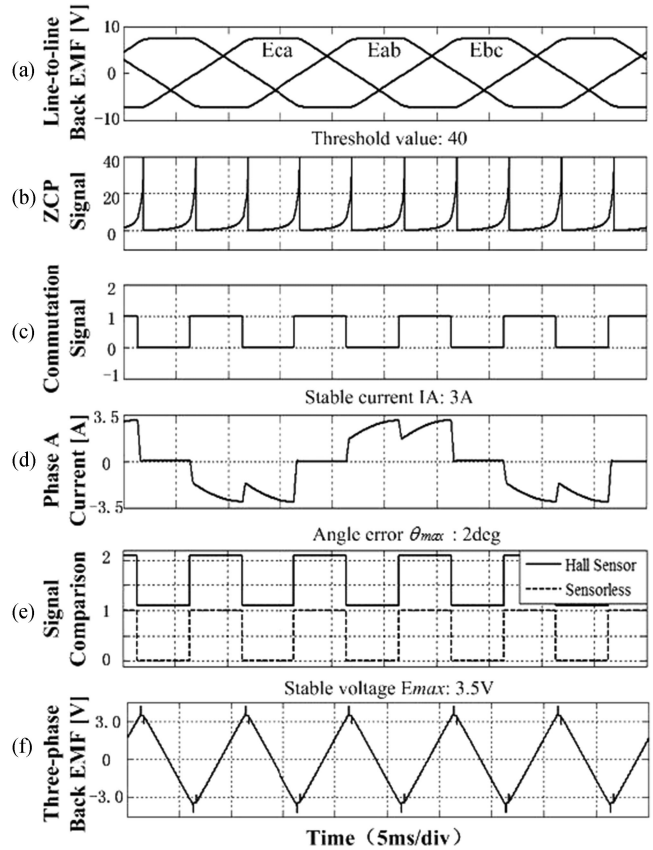

Fig. 7. Simulation results at 500 r/min. (a) Line-to-line back EMF. (b) ZCP signal. (c) Actual commutation signal. (d) Phase A current. (e) Hall sensor signal and actual commutation signal. (f) Corrected three-phase back EMF.

According to (26), a PI controller is presented to implement the closed-loop compensation. The compensation algorithm is expressed as

$$\Delta\theta_2(n) = K_p \Delta E(n) + K_i I(n) \tag{27}$$

$$I(n) = I(n-1) + \Delta E(n) \tag{28}$$

where K_p is the proportional coefficient, K_i is the integral coefficient, I is the integration difference, and n is the number of sampling.

However, due to the existence of phase inductance, the unexcited phase current cannot decrease to zero instantaneously after commutation. The measured terminal voltages generate a fluctuation simultaneously. The conventional methods neglect this voltage fluctuation because the fluctuation is slight at the high-speed motors whose phase inductance is very small. However, the motors with big phase inductance generate heavy fluctuation when the load or speed increases, and the fluctuation is shown in Fig. 4(a). The fluctuation causes a serious influence

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TABLE III
MOTOR PARAMETERS (LINE-TO-LINE)

Rated power	100W	Resistance, R	0.75 Ω
Rated DC voltage	36V	Inductance, L	2mH
Rated current, I	4.6A	Back EMF constant, K_e	5.35mV/rpm
Rated speed, n	3000rpm	Torque constant, K_t	0.0712N.m/A
Number of pole pairs, P	4	Moment of inertia, J	0.0588kg.cm^2

on compensation algorithm. According to this, a correction algorithm is proposed to eliminate the fluctuation.

The period of fluctuation t_c can be obtained by detecting the measured terminal voltages. To correct the fluctuation, a new function is defined as

$$\begin{cases} E_{\rm cor} = {\rm sgn}(\theta) \cdot E_{\max}\left(1 - \frac{t_c}{T_{60}}\right) \\ E_{\max} = \frac{2}{3} k_e \omega \end{cases} \tag{29}$$

where $E_{\rm cor}$ is the value of correction, $E_{\max}$ is the amplitude of the three-phase back EMF, T_{60} is the interval of adjacent commutation, and ${\rm sgn}(\theta)$ is a sign function and defined as

$${\rm sgn}(\theta) = \begin{cases} 1,\ \theta \in NE \\ -1,\ \theta \in PE \end{cases} \tag{30}$$

where θ is the phase angle.

The corrected three-phase back EMF is shown in Fig. 4(b).

Fig. 5 shows the proposed speed control system with open-loop and closed-loop compensation algorithms.

IV. SIMULATION AND EXPERIMENTAL RESULTS

A. Simulation Results

The proposed position-sensorless control method has been successfully verified by numerical simulation based on MATLAB Simulink. Figs. 5 and 6 show the performance of the proposed method at 50 and 500 r/min, respectively.

From Figs. 6(b) and 7(b), we can find that the ZCPs of line-to-line back EMF are detected effectively using the G function at low speed. But the ZCP signal is not consistent on commutation signal because of the angle errors. As shown in Figs. 6(e) and 7(e), the commutation error is less than 5°, and it decreases with the increase in speed. Figs. 6(f) and 7(f) indicate the fluctuation of three-phase back EMF, which increases with speed, and the load is corrected successfully. In order to further verify the stability of motor, the 0.25 N·m torque is applied, the current of phase A is shown in Fig. 7(d).

B. Experimental Results

The parameters of experimental BLDC motor are shown in Table III. The control system consists of a fixed-point digital signal processor, TMS320F2405, and a drive chip, DRV8312. The experimental platform is shown in Fig. 8.

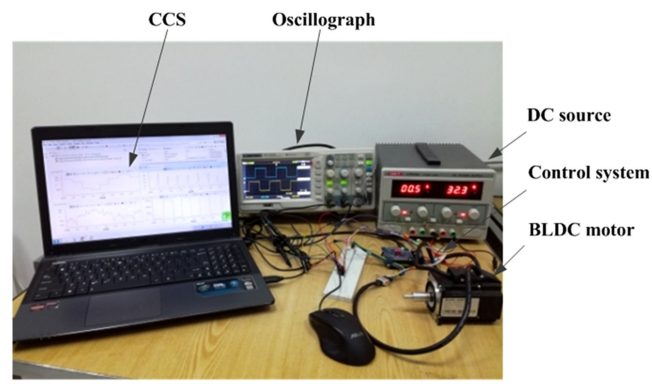

Fig. 8. Experimental platform.

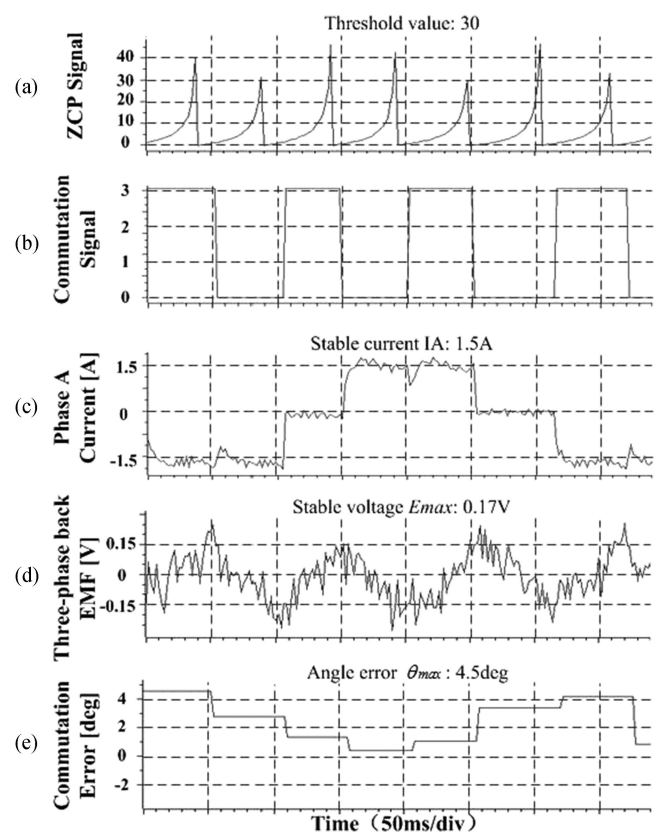

Fig. 9. Experimental results at 50 r/min. (a) ZCP signal. (b) Actual commutation signal. (c) Phase A current. (d) Corrected three-phase back EMF. (e) Commutation error.

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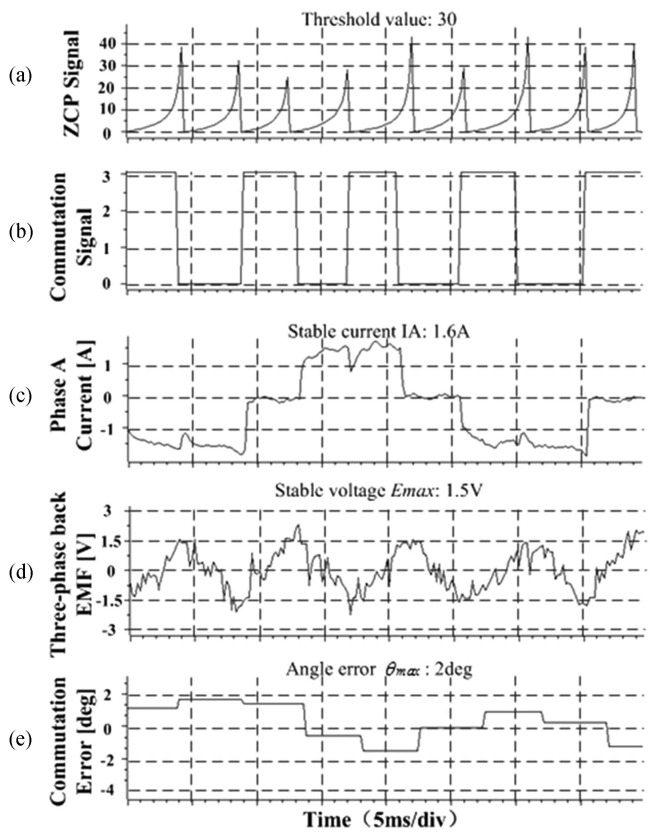

Fig. 10. Experimental results at 500 r/min. (a) ZCP signal. (b) Actual commutation signal. (c) Phase A current. (d) Corrected three-phase back EMF. (e) Commutation error.

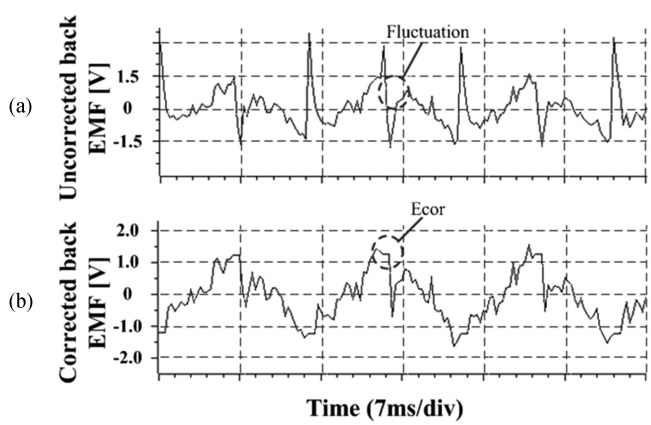

Fig. 11. Three-phase back EMF correction at 400 r/min. (a) Uncorrected. (b) Corrected.

As shown in Figs. 9(a) and 10(a), the noise of ZCP signal is very small in a low-speed range. But, due to the limitation of code composer studio graphics sampling frequency, the ZCP signal cannot be shown entirely. So, the peak points of ZCP signal are not equal. From Figs. 9(b) and 10(b), we can see that the ZCP signal, which is delayed by the LPF, has reached the preset threshold value.

From Fig. 11(a), we can see that the fluctuation of three-phase back EMF is heavy and makes the back EMF asymmetrical. But through the proposed correction algorithm, the fluctuation

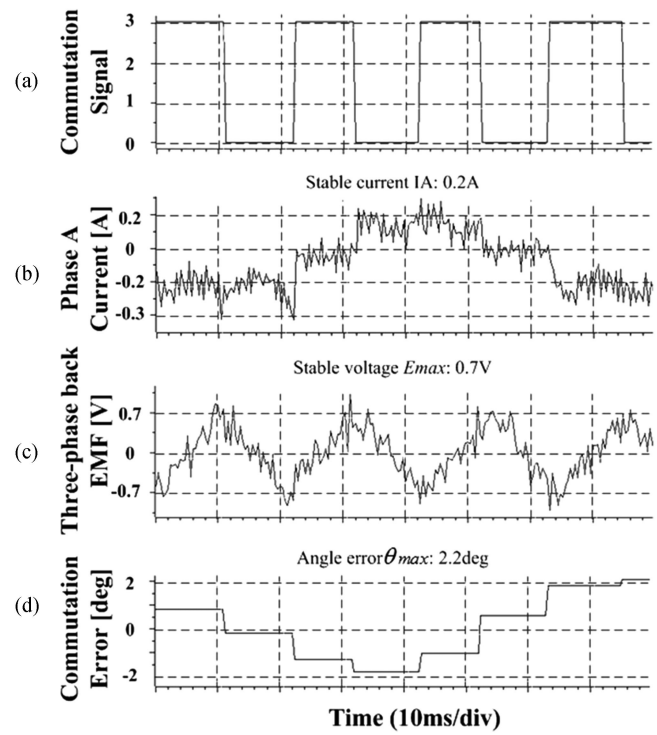

Fig. 12. Experimental results under no load at 250 r/min. (a) Commutation signal. (b) Phase A current. (c) Corrected three-phase back EMF. (d) Commutation error.

is corrected effectively, as shown in Fig. 11(b). The residual error, which is caused by measure, has only a slight influence on angle compensation.

In order to get the accuracy of the proposed position-sensorless control method, the real commutation signal is compared with the Hall sensor signal, as shown in Figs. 9(e) and 10(e). It is noted that the maximum commutation error is 5° at 50 r/min and 3° at 500 r/min. The two main reasons why there still exists commutation error are as follows.

1) The calculated speed is discontinuous and has some retard yet. So, the deviations of LPF's cutoff frequency, the corrected three-phase back EMF, and the speed PI controller are introduced.
2) The load fluctuation has an evident influence on speed and makes the speed unstable at low speed.

In order to test the stability of the motor, the comparative experiment with load and without load is illustrated, as shown in Figs. 12 and 13. Fig 12 shows the experimental results when the motor has no load. Both phase current and fluctuation of phase back EMF are very less. Without load, the commutation error tends to be stable and is less than 2°. Fig. 13 shows the experimental results under load, and the phase current and fluctuation of phase back EMF increase sharply. The fluctuation of phase back EMF is corrected successfully, and the amplitude of phase back EMF under load is nearly consistent with that under no load. From Fig. 13(b), we can see that the phase current is close to the rated current of the motor when a load is applied. Comparing Figs. 12(d) and 13(d), the fluctuation of commutation error increases obviously under the load, but the fluctuation

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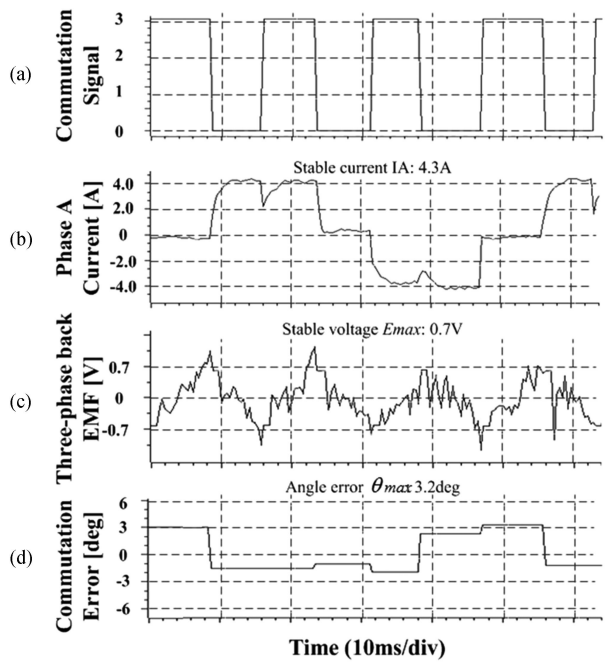

Fig. 13. Experimental results under load at 250 r/min. (a) Commutation signal. (b) Phase A current. (c) Corrected three-phase back EMF. (d) Commutation error.

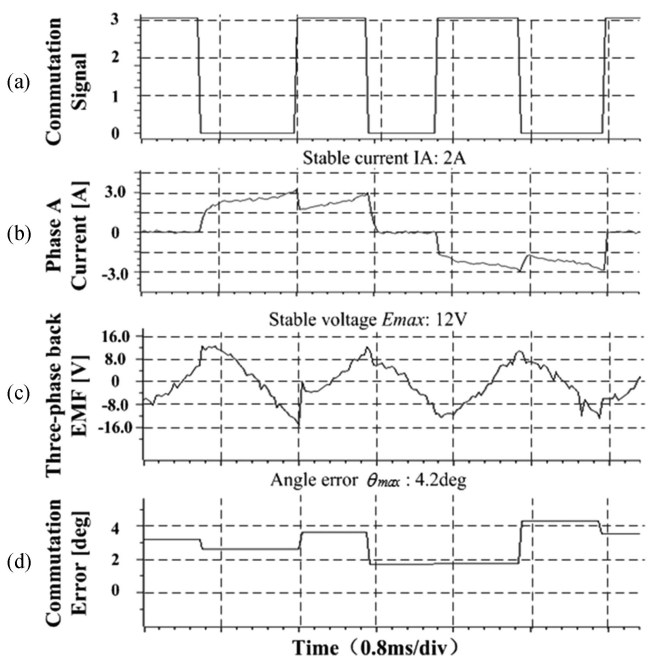

Fig. 14. Experimental results under the rated speed. (a) Commutation signal. (b) Phase A current. (c) Corrected three-phase back EMF. (d) Commutation error.

ranges are similar. Therefore, the stability of motor under load is verified successfully.

In order to verify the effectiveness of the proposed method under the rated speed, the experiment results under the rated speed are shown in Fig. 14. From Fig. 14(d), we can see that the commutation error is less than 5°. Furthermore, due to the existence of inductance, the phase current has a distinct rising

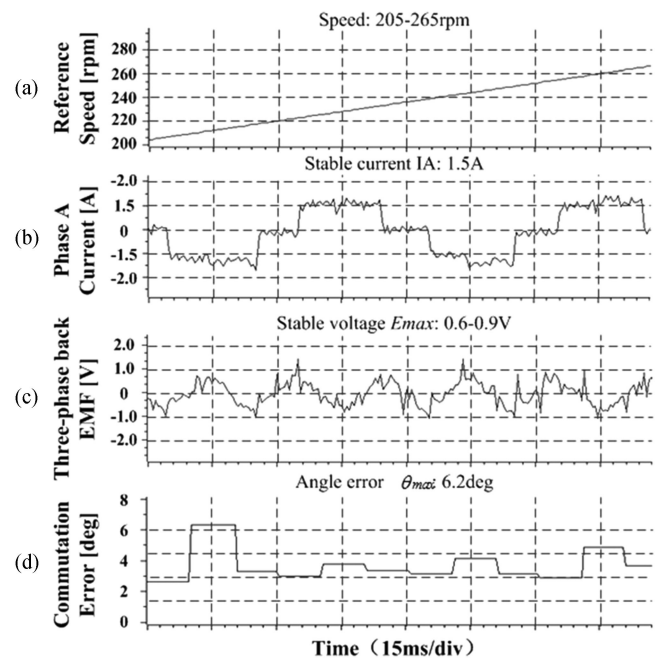

Fig. 15. Experimental results during speed change. (a) Reference speed. (b) Phase A current. (c) Corrected three-phase back EMF. (d) Commutation error.

process at high speed, which is shown in Fig. 14(b). Fig. 15 shows the dynamic performance of the proposed method. From Fig. 15(a), we can find that the reference speed increases from 205 to 265 r/min and the rate of change in speed is 500 r/min/s. The commutation error is slightly larger than that under stable speed, as shown in Fig. 15(d), but the error is allowable because the phase current shown in Fig. 15(b) illustrates the motor is running steadily.

V. Conclusion

This paper proposed a novel position-sensorless control method for BLDC motors at low speed. This method includes three new algorithms: a novel ZCP detection algorithm, which is combined G function and a digital LPF with alterable cutoff frequency; a new speed calculation algorithm; and a novel compensation algorithm. The simulation and experimental results show that the commutation error is less than 1.5% at around 1.7%–17% of the rated speed. Therefore, the stability and reliability are verified.

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